STATISTICAL LEARNING COURSEWORK | R

Breast Cancer Classification

Interpretable logistic regression, principal components, and threshold-sensitive evaluation on a public benchmark.

9
cytology features

70 / 30
stratified holdout split

5-fold
cross-validation recorded

Objective

Study how cytological features separate benign from malignant cases, compare a full-feature logistic model with a PCA-based model, and make the false-negative tradeoff visible across classification thresholds.

Methods and Report Rationale

The R workflow loads `BreastCancer` from `mlbench`, renames selected columns, converts `?` to missing values, removes incomplete records, converts nine ordinal cytology features to numeric, and encodes malignant cases as 1. It uses seed 7435 for a stratified 70/30 split.

PCA is calculated on the training features using centering and scaling. The report gives two rationales: scaling gives each feature unit variance so no single variable dominates PCA, and stratification preserves the class distribution across the split. The analysis inspects eigenvalues, variance explained, scree shape, four loading vectors, and a PC1/PC2 plot. It then fits logistic regression using all original features and, separately, the first six principal components. The full-feature model is evaluated at three probability thresholds. A bonus section fits logistic regression in five-fold cross-validation using ROC as the training metric.

Recorded Full-Feature Holdout Results

Probability cutoff	Accuracy	Sensitivity	False negatives
0.3	95.10%	95.35%	4
0.5	93.63%	91.86%	7
0.7	91.67%	86.05%	12

The report identifies false negatives (malignant cases predicted as benign) as the most concerning error. Its scree interpretation says the elbow suggests one or two components, while the 90% cumulative-variance rule requires six. The variance table reports 65.53% for PC1, 8.74% for PC2, and 93.29% cumulatively for PC1 through PC6.

PCA Logistic Model

The report narrative states that the six-component PCA model achieves 94.12% holdout accuracy. The displayed PCA accuracy output is 0.9363 (93.63%); the report also displays 91.86% sensitivity, 94.92% specificity, and 92.94% precision.

Five-Fold Cross-Validation

The bonus output reports 96.78% accuracy, 95.40% sensitivity, 97.52% specificity, and 0.9292 Kappa. The report describes this model as superior to the single holdout split across the listed evaluation metrics.

Reproduction

The R code is arranged in `analysis.R` in the same sequence as the original HTML report.

Summary based on the original Breast-Cancer-Analysis.R.html project report.